



Virtual Internship

SQL Query Debugging

Week 2 **Task 2**



Bug



Fix

**Query 1 of 9**

How many unique customers are in the city of 'Surat'?

The Bug

```
SELECT
    COUNT(DISTINCT customer_id) AS
distinct_customers
FORM gdb080.dim_customers
WHERE city == 'surat';
```

The Fix

```
SELECT
    DISTINCT(COUNT(customer_id)) AS
unique_customers
FROM gdb080.dim_customers
WHERE city = "Surat";
```

- FORM was a Typo for FROM
- MYSQL Does not support == , use = or LIKE



Query 2 of 9

What are the min and max order quantities for each product?

The Bug

```
SELECT
  p.product_id,
  p.product_name,
  MIN(p.order_qty) as minimum_qty
  MAX(p.order_qty) as maximum_qty
FROM gdb080.fact_order_lines f
JOIN gdb080.dim_products p ON f.product_id =
p.product_id
GROUP BY p.product_id;
```

The Fix

```
SELECT
  dp.product_id,
  dp.product_name,
  MIN(fol.order_qty) AS min_order_qty,
  MAX(fol.order_qty) AS max_order_qty
FROM fact_order_lines fol
JOIN dim_products dp
ON dp.product_id = fol.product_id
GROUP BY dp.product_id;
```

- Missing Comma in SELECT LIST
- Order Qty lives in Fact tables not in the Product tables



Query 3 of 9

Unfulfilled_orders(i.e order_qty - delivery_qty) by month

The Bug

```
SELECT
    MONTHNAME(order_placement_date as month_name,
    SUM(order_qty-delivery_qty)
FROM gdb080.fact_orders_lines
GROUP By MONTHNAME(order_placement_date)
ORDER BY unfulfilled_orders DESC;
```

The Fix

```
SELECT
    MONTHNAME(order_placement_date) as month_name,
    SUM(order_qty-delivery_qty) AS unfulfilled_orders
FROM gdb080.fact_order_lines
GROUP By MONTHNAME(order_placement_date)
ORDER BY unfulfilled_orders DESC;
```

- Closing Bracket was in wrong place
- Gaved the SUM an alias so ORDER BY can find it



Query 4 of 9

The percentage breakdown of order_qty by category

The Bug

```
with total_order_qty_by_category as
(
    SELECT
    p.category,
    SUM(f.order_qty) as total_quantity
    FROM gdb080.dim_products p
    JOIN gdb080.fact_order_lines f ON p.product_id = f.product_id
    GROUP BY p.category ;
)
SELECT
category,
ROUND(100 * total_quantity / SUM(total_quantity) OVER (), 2) AS
order_qty_pct
FROM total_order_quantity_by_category
order by order_qty_pct DESC;
```



Query 4 of 9

The percentage breakdown of order_qty by category

The Fix

```
WITH total_order_qty_by_category AS
(
    SELECT
        p.category,
        SUM(f.order_qty) AS total_quantity
    FROM dim_products p
    JOIN fact_order_lines f ON p.product_id = f.product_id
    GROUP BY p.category
)
SELECT
    category,
    ROUND(100 * total_quantity / SUM(total_quantity)
    OVER (), 2) AS order_qty_pct
FROM total_order_qty_by_category
ORDER BY order_qty_pct DESC ;
```

-- Semi Colon inside the CTE ended the query early
-- CTE name didn't match both the sides



Query 5 of 9

Customer on time target % and percentage category

The Bug

```
SELECT
    customer_id,
    customer_name,
    t.ontime_target_pct,
    CASE
        WHEN t.ontime_target_pct > 90 THEN 'Above 90'
        WHEN t.ontime_target_pct > 80 THEN 'Above 80'
        WHEN t.ontime_target_pct > 70 THEN 'Above 70'
        ELSE "Below 70"
    END ASpercentage_category ,
FROM gdb080.dim_targets_orders t
JOIN gdb080.dim_customers c
ON t.customer_id = c.customer_id;
```




Query 5 of 9

Customer on time target % and percentage category

The Fix

```
SELECT
    c.customer_id,
    c.customer_name,
    t.ontime_target_pct,
    CASE
        WHEN t.ontime_target_pct > 90 THEN 'Above 90'
        WHEN t.ontime_target_pct > 80 THEN 'Above 80'
        WHEN t.ontime_target_pct > 70 THEN 'Above 70'
        ELSE 'Below 70'
    END AS percentage_category
FROM gdb080.dim_targets_orders t
JOIN gdb080.dim_customers c
ON t.customer_id = c.customer_id;
```

-- Quotation mark didn't match

-- customer_id, customer_name need a table prefix , trailing comma removed



Query 6 of 9

Product categories with product name and count per category

The Bug

```
SELECT category, GROUP_CONCAT(product_name)
AS products COUNT(*) AS product_count
FROM gdb080.dim_products
GROUP category;
```

The Fix

```
SELECT category, GROUP_CONCAT(product_name) AS
products ,
COUNT(*) AS product_count
FROM gdb080.dim_products
GROUP BY category;
```

- Missing comma between two columns
- GROUP BY needs BY word ,GROUP Alone doesn't work



Query 7 of 9

Top 3 most demanded products in the 'Dairy' category(Order_qty in Mlns)

The Bug

```
SELECT
p.product_name,
ROUND(SUM(f.order_qt) / 1000000,2) AS order_qty_mln
FROM gdb080.dim_products p
JOIN gdb080.fact_order_lines f
WHERE p.category = 'Dairy'
GROUP BY p.product_name
ORDER BY order_qty_mln DESC
LIMIT 3;
```

The Fix

```
SELECT
    p.product_name,
    ROUND(SUM(f.order_qty) / 1000000,2) AS order_qty_mln
FROM gdb080.dim_products p
JOIN gdb080.fact_order_lines f
ON p.product_id = f.product_id
WHERE p.category = 'Dairy'
GROUP BY p.product_name
ORDER BY order_qty_mln DESC
LIMIT 3;
```

-- Order_qt missing a letter

-- JOIN needs an ON to know how to connect the tables



Query 8 of 9

OTIF % for customer “Vijay Stores”

The Bug

```
SELECT
  c.customer_names,
  ROUND((SUM(f.otif) / COUNT(f.order_id) * 100),2) AS
  OTIF_percentage
FROM gdb080.fact_orders_aggregate f
JOIN gdb080.dim_customers c
ON c.customer_id = f.customer_id
GROUP BY c.customer_name
WHERE c.customer_name = "Vijay Stores";
```

The Fix

```
SELECT
  c.customer_name,
  ROUND((SUM(f.otif) / COUNT(f.order_id) * 100),2) AS
  OTIF_percentage
FROM gdb080.fact_orders_aggregate f
JOIN gdb080.dim_customers c
ON c.customer_id = f.customer_id
WHERE c.customer_name = "Vijay Stores"
GROUP BY c.customer_name;
```

- customer_name column has added extra letter
- WHERE clause must come before GROUP BY



Query 9 of 9

In Full % per product and which product ranks highest

The Bug

```
WITH product_if_target AS (  
    SELECT  
        p.product_name,  
        SUM(CASE WHEN f.in_full = 1 THEN 1 ELSE 0) AS if_count,  
        COUNT(f.order_id) AS total_count  
    FROM gdb080.fact_order_lines f  
    JOIN gdb080.dim_products p ON p.product_id = f.product_id  
    GROUP BY p.product_name  
)  
SELECT  
    product_name,  
    ROUND(if_count / total_count * 100), 2 AS  
IF_percentage  
FROM product_if_target  
order by IF_percentage DESC
```



Query 9 of 9

In Full % per product and which product ranks highest

The Fix

```
WITH product_if_target AS (  
    SELECT  
        p.product_name,  
        SUM(CASE WHEN f.in_full = 1 THEN 1 ELSE 0 END ) AS  
if_count,  
        COUNT(f.order_id) AS total_count  
    FROM  
        gdb080.fact_order_lines f  
        JOIN gdb080.dim_products p ON p.product_id =  
f.product_id  
    GROUP BY p.product_name  
)  
SELECT  
    product_name,  
    ROUND(if_count / total_count * 100, 2) AS IF_percentage  
FROM  
    product_if_target  
order by IF_percentage DESC;
```

- CASE statement was missing with END
- One extra bracket broken the ROUND function

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**Most SQL bugs are not
always about logic**

**They Hide small syntax
details**

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